



What is driving high-frequency FX?

Main highlights: US equities extended their rally over the past week, recovering from the volatility triggered by earlier geopolitical tensions. Against this backdrop, our high-frequency analysis shows a further strengthening in US equity–FX connectivity, with equity markets now driving more than 60% of the currency pairs in our sample.

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Broader cross-asset drivers: The influence of US rates on FX dynamics also increased, underscoring their growing importance as a cross-asset driver. Oil's contribution to currency moves was broadly unchanged, while EM equities and copper saw a modest increase in influence, pointing to a slight broadening in the set of risk and commodity factors shaping FX performance.

Impact across individual FX pairs: US equities remain the dominant driver of USD/JPY, AUD/USD and USD/SGD. For EUR/USD, USD/CAD and USD/CNH, high-frequency moves are driven primarily by a combination of US equities and US rates, highlighting the joint importance of risk sentiment and interest-rate dynamics. Elsewhere, oil remains the key influence on EUR/NOK, consistent with the pair's close relationship with energy markets. USD/CHF is influenced by a broader mix of factors, with US equities, US rates and copper all playing important roles.

Historical impact over the past three months: US equities have been the dominant driver of USD/CHF, USD/SGD and NZD/USD, explaining market moves on more than 90% of trading days. US rates have been especially influential for NZD/USD and USD/CNH, accounting for more than 60% of trading days. EM equities have had the strongest impact on USD/CAD, EUR/NOK and USD/MXN, contributing on more than 20% of trading days. Oil remains the defining factor for EUR/NOK, driving the pair on more than 90% of trading days.

Correlation structure: Underlying cross-asset correlations continue to show a classic risk-on/risk-off pattern. US and EM equities are generally positively correlated with pro-cyclical assets, including AUD, NZD, GBP, EUR, SGD, MXN, ZAR, gold (XAU) and bitcoin (XBT), reflecting their shared sensitivity to global growth and investor risk appetite. By contrast, oil is negatively correlated with several currencies against the US dollar, notably ZAR, SGD, CHF, CNH, NZD, XAU and AUD. US rates also maintain broadly negative correlations with major currencies, with the strongest inverse relationships in SGD, ZAR, JPY, CHF, NZD, AUD, XAU, EUR and GBP.

Network view: Beyond predictive linkages, the correlation-based Minimum Spanning Tree (MST) shows how contemporaneous cross-market relationships are structured. The network continues to identify US interest rates as the primary hub in high-frequency FX dynamics, with the VIX remaining the second most important source of cross-asset connectivity. Within FX, USD/ZAR retains its role as a key bridge currency, linking rates, equities and commodities. Its central



position highlights its role as a key connector across asset classes, making it an important channel for cross-market spillovers.

Figure 1, Figure 3 and Figure 15 look at causality in FX. Figure 1 shows the currencies whose moves can be statistically predicted by other asset classes. Figure 3 ranks the drivers of FX (highest to lowest number of currencies, driven by a given asset). Figure 15 shows the relative importance of each asset in driving a given currency in the recent period. Figure 13 looks at contemporaneous correlations. We show the top 3 correlations between FX pairs and other asset classes. Figure 14 shows intra-FX correlations. We use Granger causality tests to measure causality. All statistics measured at 5-minute frequency. For further details, please see our updated [methodology](#). For a summary of our [FX quant tools](#), please also see [DB FX research quant tools](#).

Figure 1: Significant connections from assets to currencies

Table 1: Significant connections from assets to currencies

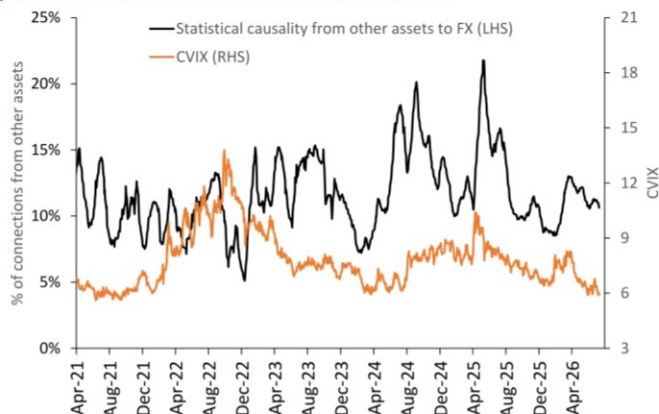
	# of conn. from other assets (Over last 4 weeks)	Assets driving currencies (Over last 4 weeks)	%tile rank	
			6M history	Since '16
Highest ↑ Connections from other assets ↓ Lowest	4	US EQ,US RATE,OIL,EM EQ	97%	94%
	4	EU EQ,EU RATE,US EQ,US RATE	81%	92%
	4	AU EQ,US EQ,COPPER,VIX	31%	46%
	3	US EQ,US RATE,EM EQ	83%	78%
	3	US EQ,US RATE,COPPER	86%	69%
	3	EU EQ,US RATE,OIL	80%	72%
	2	US EQ,US RATE	34%	46%
	2	US EQ,US RATE	66%	51%
	2	US EQ,US RATE	51%	53%
	2	US EQ,OIL	96%	60%
	1	US EQ	12%	25%
	1	US EQ	22%	26%
	1	EM EQ	19%	26%
	1	US EQ	10%	21%
	1	OIL	0%	9%
	1	COPPER	41%	65%

Max no. of connections = 15

Source: Deutsche Bank Research

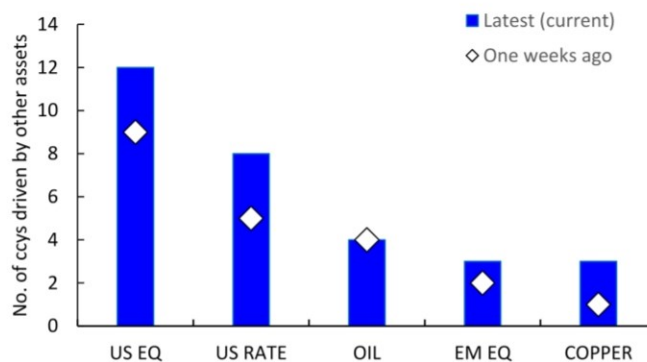


Figure 2: Extent to which other markets are driving FX (connections from other assets to FX)



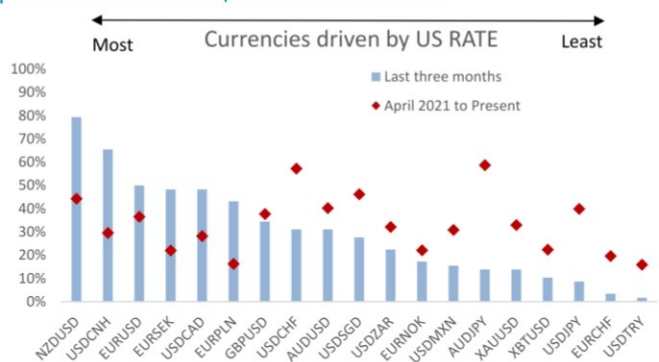
Source: Deutsche Bank, EBS, Reuters; 4-week rolling average connections

Figure 3: Number of currencies driven by each asset



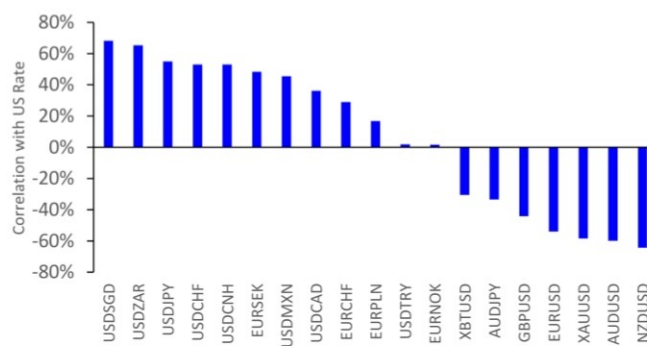
Source: Deutsche Bank, EBS, Reuters; Total number of currencies = 19

Figure 4: NZD/USD and USD/CNH are most driven by US rates over the past three months



Source: Deutsche Bank Research

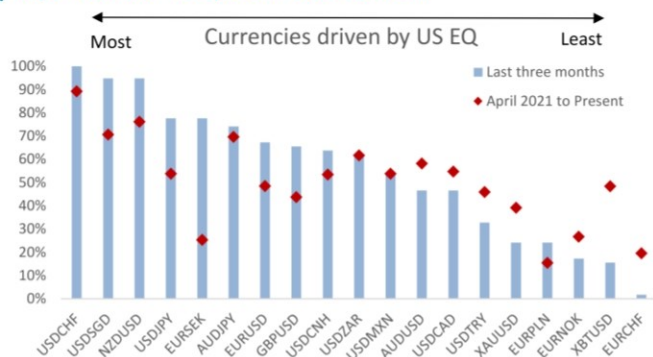
Figure 5: Correlation with US rates



Source: Deutsche Bank Research

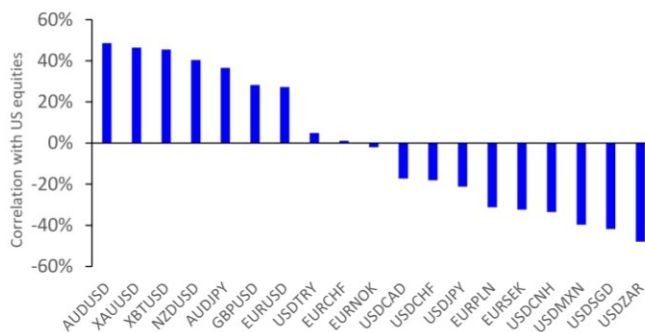


Figure 6: USD/CHF and USD/SGD most driven by US equities over the past three months



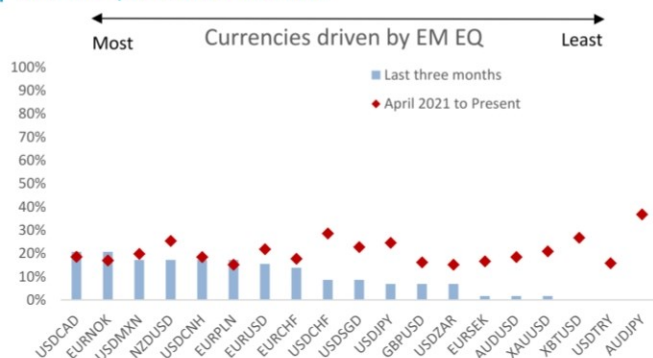
Source: Deutsche Bank Research

Figure 7: Correlation with US equities



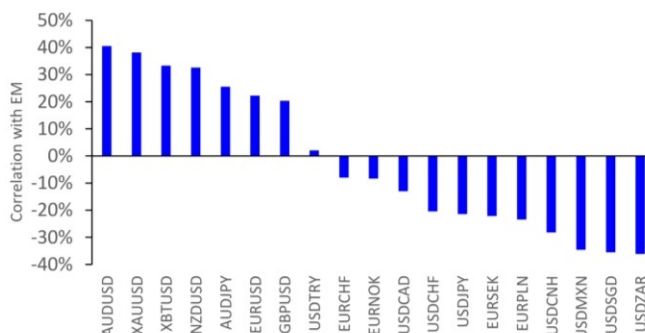
Source: Deutsche Bank Research

Figure 8: USD/CAD and EUR/NOK most driven by EM equities over the past three months



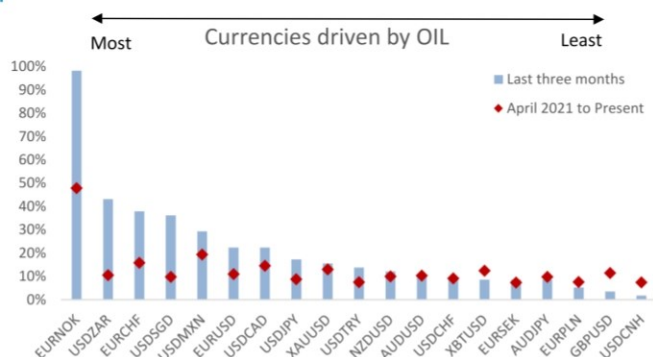
Source: Deutsche Bank Research

Figure 9: Correlation with EM equities



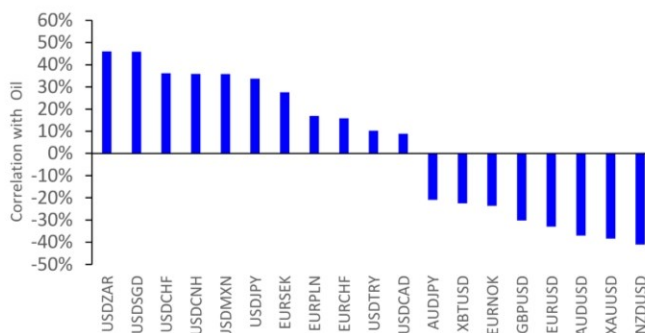
Source: Deutsche Bank Research

Figure 10: EUR/NOK most driven by Oil over the past three months



Source: Deutsche Bank Research

Figure 11: Correlation with Oil



Source: Deutsche Bank Research

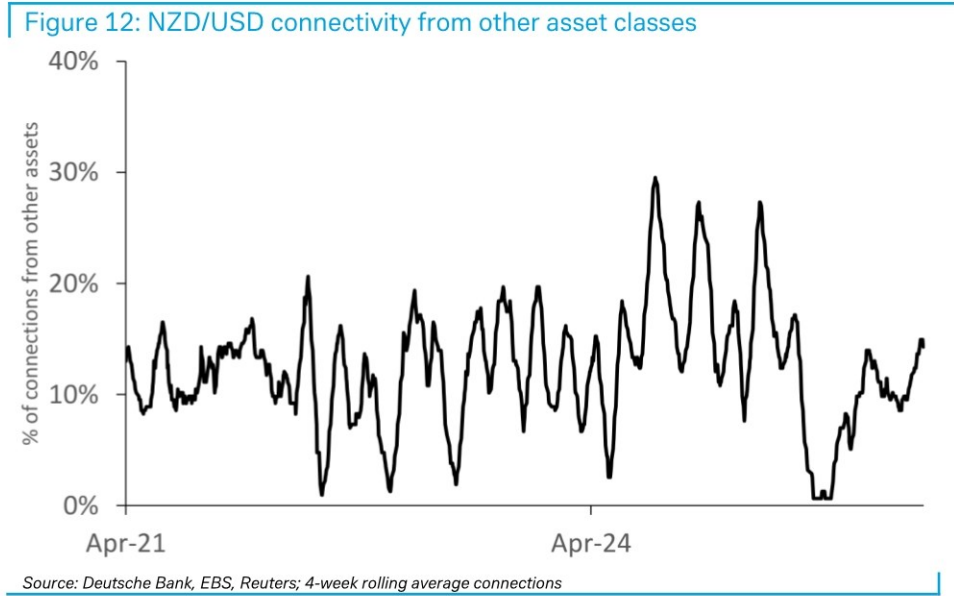




Figure 13: Cross-asset correlation

▲ ▼ Up (stronger correlation)/down (weaker correlation) against last week/month 5-day correlations

5 Day correlations	First			Second			Third		
	Current	1 week ago	1 month ago	Current	1 week ago	1 month ago	Current	1 week ago	1 month ago
EURUSD	US RATE	▲	▲	VIX	▲	▼	COPPER	▼	▼
	-54%	-47%	-49%	-34%	-19%	-41%	34%	37%	45%
USDJPY	US RATE	▲	▲	EU RATE	▲	▲	UK RATE	▲	▲
	55%	49%	32%	44%	31%	35%	44%	33%	32%
GBPUSD	US RATE	▲	▼	VIX	▲	▼	COPPER	▼	▼
	-44%	-43%	-51%	-37%	-30%	-40%	31%	36%	46%
EURNOK	OIL	▼	▼	VIX	▼	▼	EM EQ	▼	▲
	-24%	-25%	-24%	11%	26%	14%	-8%	-11%	3%
EURSEK	US RATE	▲	▲	VIX	▲	▲	EU RATE	▲	▲
	48%	22%	39%	47%	36%	39%	37%	29%	36%
USDCHF	US RATE	▲	▲	UK RATE	▲	▼	EU RATE	▲	▼
	53%	49%	52%	40%	31%	53%	40%	36%	53%
EURCHF	US RATE	▲	▼	EU RATE	▼	▼	UK RATE	▲	▲
	29%	26%	32%	25%	26%	28%	25%	21%	25%
AUDJPY	VIX	▲	▲	COPPER	▲	▲	US EQ	▲	▲
	-50%	-29%	-30%	37%	19%	30%	37%	26%	29%
AUDUSD	US RATE	▲	▲	VIX	▲	▼	COPPER	▲	▼
	-60%	-45%	-50%	-54%	-39%	-61%	52%	44%	59%
NZDUSD	US RATE	▲	▲	EU RATE	▲	▼	VIX	▲	▼
	-64%	-50%	-42%	-49%	-43%	-57%	-47%	-38%	-52%
USDCAD	US RATE	▼	▲	VIX	▲	▼	COPPER	▼	▼
	36%	36%	15%	26%	22%	28%	-21%	-38%	-33%
XAUUSD	US RATE	▲	▲	COPPER	▲	▼	US EQ	▲	▼
	-58%	-47%	-43%	57%	55%	62%	46%	36%	48%
USDCNH	US RATE	▲	▲	VIX	▲	▲	UK RATE	▲	▼
	53%	43%	50%	50%	31%	44%	39%	32%	52%
USDSGD	US RATE	▲	▲	VIX	▲	▼	EU RATE	▲	▼
	68%	53%	51%	51%	31%	54%	51%	38%	55%
USDMXN	VIX	▲	▲	US RATE	▼	▲	US EQ	▲	▼
	49%	36%	48%	46%	49%	39%	-40%	-34%	-46%
USDZAR	US RATE	▲	▲	VIX	▲	▼	EU RATE	▲	▼
	65%	52%	48%	56%	39%	57%	48%	42%	53%
USDTRY	OIL	▲	▲	EU RATE	▼	▲	SZ EQ	▲	▲
	10%	0%	-2%	9%	10%	-5%	-6%	-4%	3%
EURPLN	US EQ	▲	▼	VIX	▲	▼	COPPER	▲	▼
	-31%	-17%	-37%	31%	26%	48%	-25%	-19%	-29%
XBTUSD	VIX	▲	▲	US EQ	▲	▼	EM EQ	▲	▲
	-52%	-45%	-48%	46%	44%	49%	33%	31%	12%

Source: Deutsche Bank Research, EBS, Reuters We compute daily correlations using log price changes sampled at 5 minutes frequency and take an average of the daily correlations over the past five days to arrive at the final 5-day correlation numbers.



Figure 14: Cross-currency correlation

▲ Up (stronger correlation)/down (weaker correlation) against last month 5-day correlations

5 Day correlations	EURUSD	USDJPY	GBPUSD	USDCHF	AUDUSD	NZDUSD	USDCAD	XAUUSD	USDCNH	USDSGD	USDMXN	USDTRY	USDZAR	XBTUSD
Current	USDSGD ▲	USDSGD ▲	EURUSD ▼	EURUSD ▼	NZDUSD ▲	AUDUSD ▲	AUDUSD ▲	USDZAR ▼	USDSGD ▲	AUDUSD ▼	USDZAR ▼	EURSEK ▲	USDSGD ▲	XAUUSD ▲
1 month ago	-82%	72%	79%	-76%	89%	89%	-63%	-61%	72%	-85%	67%	5%	76%	35%
Current	NZDUSD ▲	NZDUSD ▲	AUDUSD ▼	NZDUSD ▼	USDSGD ▼	USDSGD ▲	NZDUSD ▲	USDSGD ▼	AUDUSD ▼	NZDUSD ▲	AUDUSD ▼	GBPUSD ▲	AUDUSD ▼	USDZAR ▼
1 month ago	80%	-63%	70%	-69%	-85%	-84%	-62%	-60%	-69%	-84%	-66%	-5%	-73%	-32%
Current	GBPUSD ▼	EURUSD ▲	USDSGD ▼	GBPUSD ▼	EURUSD ▼	EURUSD ▼	USDCHF ▲	AUDUSD ▼	NZDUSD ▲	EURUSD ▲	USDSGD ▼	AUDUSD ▼	NZDUSD ▼	USDSGD ▼
1 month ago	75%	-62%	77%	-74%	-87%	-71%	-49%	-61%	-69%	-71%	-72%	1%	-77%	-36%
Current	79%	-63%	-69%	-68%	78%	80%	59%	58%	-66%	-82%	65%	4%	-68%	-30%
1 month ago	85%	-59%	-75%	-74%	80%	75%	43%	66%	-65%	-82%	68%	0%	-70%	-34%
Current	AUDUSD ▼	AUDUSD ▲	NZDUSD ▼	USDSGD ▼	USDZAR ▼	GBPUSD ▼	EURUSD ▼	NZDUSD ▼	EURUSD ▼	USDZAR ▼	NZDUSD ▲	EURCHF ▲	USDMXN ▼	AUDUSD ▼
1 month ago	78%	-59%	69%	68%	-73%	69%	-59%	54%	-65%	76%	-62%	4%	67%	29%
Current	80%	-55%	75%	68%	-77%	75%	-58%	58%	-66%	75%	-59%	0%	69%	37%
1 month ago	USDCHF ▼	USDCHF ▲	USDCHF ▼	AUDUSD ▼	GBPUSD ▼	USDCHF ▼	GBPUSD ▲	USDMXN ▼	GBPUSD ▼	USDJPY ▲	EURUSD ▼	USDZAR ▼	XAUUSD ▼	USDMXN ▼
Current	-76%	57%	-68%	-66%	70%	-69%	-57%	-47%	-56%	72%	-60%	-4%	-61%	-26%
1 month ago	-77%	56%	-74%	-68%	77%	-74%	-54%	-54%	-63%	60%	-67%	-1%	-61%	-32%
Current	USDCNH	GBPUSD ▼	USDCAD ▲	EURCHF ▲	USDCNH ▼	USDZAR ▼	USDSGD ▲	AUDJPY ▼	USDZAR ▼	USDCNH ▲	GBPUSD ▼	EURPLN ▲	EURUSD ▼	NZDUSD ▼
1 month ago	-65%	-55%	-57%	59%	-69%	-68%	55%	42%	54%	72%	-53%	3%	-59%	25%
Current	USDJPY ▲	USDZAR ▲	USDCNH ▼	USDCAD ▲	USDMXN ▼	USDCNH ▲	USDJPY ▼	EURUSD ▼	USDCHF ▼	GBPUSD ▼	USDCNH ▼	USDCNH ▼	USDCNH ▼	EURSEK ▲
1 month ago	-63%	52%	-56%	59%	-66%	-66%	45%	41%	54%	-69%	52%	2%	54%	-21%
Current	USDMXN ▼	USDCNH ▼	USDJPY ▼	USDJPY ▲	USDCHF ▼	USDJPY ▲	USDCNH ▼	USDCNH ▼	USDMXN ▼	USDCHF ▼	XAUUSD ▼	EURUSD ▼	USDJPY ▲	EURPLN ▼
1 month ago	-60%	50%	-55%	57%	-66%	-63%	41%	-40%	52%	68%	-47%	-2%	52%	-20%
Current	USDZAR ▼	USDMXN ▲	USDMXN ▼	USDCNH ▼	USDCAD ▲	USDCAD ▼	USDMXN ▼	USDCHF ▼	USDJPY ▼	USDMXN ▼	USDJPY ▲	USDCAD ▼	GBPUSD ▼	USDCNH ▼
1 month ago	-59%	46%	-53%	54%	-63%	-62%	39%	-39%	50%	65%	46%	-2%	-51%	-20%
Current	USDCAD ▲	USDCAD ▲	USDZAR ▼	USDZAR ▼	AUDJPY ▼	USDMXN ▲	USDZAR ▼	GBPUSD ▼	USDCAD ▲	XAUUSD ▼	USDCHF ▼	USDCHF ▼	USDCHF ▼	AUDJPY ▼
1 month ago	-59%	45%	-51%	49%	59%	-62%	38%	38%	41%	-60%	43%	2%	49%	19%
Current	XAUUSD ▼	XAUUSD ▼	EURSEK ▼	USDMXN ▼	USDJPY ▲	XAUUSD ▼	AUDJPY ▼	USDJPY ▼	XAUUSD ▼	USDCAD ▲	USDCAD ▼	USDSGD ▲	EURSEK ▲	GBPUSD ▼
1 month ago	41%	-37%	-39%	43%	-59%	54%	-36%	-37%	-40%	55%	39%	1%	44%	17%
Current	EURSEK ▼	EURSEK ▲	XAUUSD ▼	XAUUSD ▼	XAUUSD ▼	EURSEK ▲	XAUUSD ▼	XBTUSD ▲	EURSEK ▲	EURSEK ▼	AUDJPY ▼	XAUUSD ▼	AUDJPY ▼	USDJPY ▼
1 month ago	-37%	35%	38%	-39%	58%	-50%	-29%	35%	39%	45%	-36%	1%	-39%	-15%
Current	-42%	32%	53%	-49%	66%	-46%	-40%	32%	38%	51%	-64%	4%	-60%	-14%

Highest

Lowest

Source: Deutsche Bank Research

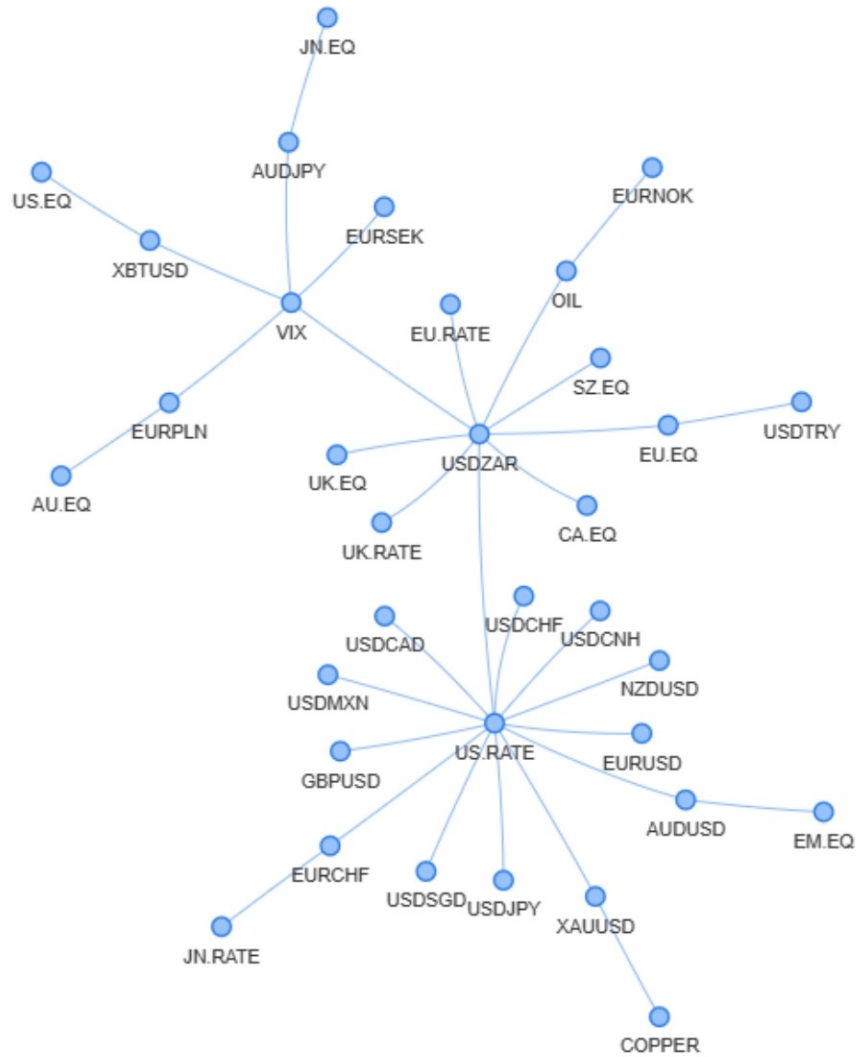


Figure 15: Currencies driven by other asset classes over time

Assets driving CCY (horizontal)	Equities			Rates		Commodities		Vol
	US	EM	EU/JN/UK/AU/CA/SZ	US	EU/JN/UK	Copper	Oil	VIX
EURUSD	48%	22%	41%	36%	22%	14%	11%	8%
USDJPY	54%	24%	7%	40%	9%	12%	9%	8%
GBPUSD	44%	16%	23%	38%	27%	10%	11%	7%
EURNOK	27%	17%	80%	22%	19%	15%	48%	8%
EURSEK	25%	17%	71%	22%	12%	13%	7%	9%
USDCHF	89%	29%	19%	57%		19%	9%	4%
EURCHF	20%	18%	63%	20%	13%	10%	16%	11%
AUDUSD	58%	18%	37%	40%		19%	10%	8%
NZDUSD	76%	25%		44%		26%	10%	11%
USDCAD	55%	19%	37%	28%		8%	14%	10%
AUDJPY	70%	37%	95%	59%	9%	36%	10%	14%
XAUUSD	39%	21%		33%		16%	13%	14%
XBTUSD	48%	27%		22%		11%	12%	11%
USDCNH	53%	18%		30%		16%	7%	12%
USDSGD	71%	23%		46%		21%	10%	10%
USDMXN	54%	20%		31%		14%	19%	17%
USDTRY	46%	16%		16%		5%	7%	4%
USDZAR	62%	15%		32%		17%	11%	13%
EURPLN	15%	15%	91%	16%	15%	14%	8%	9%

Source: Deutsche Bank Research, EBS, Reuters % of days since 1st April 2021 each currency is driven by other assets. For example, EURUSD is driven by US equities 48% of the days. The color coding is horizontal with the darkest green cell in each row representing the most important asset over time for a given currency. Equities: EU for EUR crosses, JN for USDJPY, AU for AUDUSD and AUDJPY, SZ for USDCHF, CA for USDCAD, UK for GBPUSD. Rates: EU for EUR crosses, JN for USDJPY, AUDJPY and UK for GBPUSD

Figure 16: US rates remain the most important focal point based on correlation



Source: Deutsche Bank Research



Appendix 1

Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s). In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Rohini Grover, Ph.D..

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